

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
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Date:	06-10-2026	Duration:	60 minutes

Code of Conduct

I Yoga Madha A/192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



AI-Based Emergency Response and First-Responder Coordination System

1. Understand the Problem Statement

1.1 Problem Explanation

During an emergency, quick access to accurate information and proper coordination between the patient, first responder, ambulance team, and hospital is very important. In the existing approach, emergency information is often communicated through phone calls, paper records, or separate systems. Important details such as allergies, medications, medical history, emergency contacts, and current location may not be immediately available.

This can cause delays in decision-making and emergency treatment.

The proposed **AI-Based Emergency Response and First-Responder Coordination System** provides a centralized and secure software platform that collects important emergency information, generates an AI-based emergency summary, alerts first responders about critical information, shares the patient's location with authorized users, and notifies the appropriate hospital.

1.2 Objectives

The main objectives are:

1. To provide quick access to essential emergency information.
2. To generate an AI-based summary of the patient's critical medical information.

3. To identify and highlight critical allergies, medications, and medical conditions.
4. To improve coordination between first responders, ambulance teams, and hospitals.
5. To provide real-time emergency location sharing.
6. To notify emergency contacts and hospitals when required.
7. To protect patient information using authentication and access control.
8. To reduce delays and improve emergency decision-making.

1.3 Scope

The system covers emergency information management, AI-based emergency summarization, first-responder access, location sharing, hospital notification, emergency contact management, and secure access to authorized medical information.

The system can be used by **patients, first responders, ambulance teams, hospitals, doctors, and emergency contacts**.

The initial system focuses on software-based emergency coordination. Future versions can integrate wearable devices, IoT sensors, hospital systems, and advanced AI prediction.

2. Key Stakeholders and Challenges

2.1 Stakeholders

Stakeholder	Role and Responsibility
Patient	Provides medical information, emergency contacts, allergies, medications, and consent for data access.
First Responder	Accesses authorized emergency information and provides immediate assistance.
Ambulance Team	Receives emergency details and location and coordinates transportation.
Doctor/Hospital	Receives emergency information and prepares for treatment.
Emergency Contact	Receives alerts and important emergency notifications.
System Administrator	Manages users, permissions, security, and system maintenance.
AI Module	Generates emergency summaries and identifies potentially important information.

2.2 Major Challenges

Patient Challenges

- May be unconscious or unable to communicate.
- Medical information may not be immediately available.
- Privacy of personal medical data must be protected.

First Responder Challenges

- Limited time for decision-making.
- Difficulty obtaining accurate patient information.
- Too much information can delay understanding.

Ambulance Challenges

- Need accurate location information.
- Need communication with hospitals.
- Need quick access to emergency details.

Hospital Challenges

- Need advance information about incoming patients.
- Incomplete information can delay preparation.
- Multiple emergency cases may need to be coordinated.

System Challenges

- Protecting sensitive information.
- Maintaining high availability.
- Preventing unauthorized access.
- Handling large numbers of simultaneous emergencies.

3. Analyze the Current Approach

3.1 Existing System

Currently, emergency response commonly depends on:

- Phone calls to emergency services.
- Verbal communication.
- Paper-based or physical medical records.
- Separate hospital information systems.
- Manual communication between ambulance and hospital.
- Manual sharing of emergency contact details.
- GPS/location sharing through separate applications.

The first responder may need to collect information manually from the patient, relatives, identification cards, or available records.

3.2 Strengths of Current Approach

1. Easy to understand and use.
2. Phone communication is widely available.
3. Emergency services already have established procedures.
4. Human decision-making is available.
5. Does not require sophisticated technology in every situation.

3.3 Limitations

1. Important medical information may be unavailable.
2. Communication can be slow.
3. Information may be incomplete or inaccurate.
4. Patient allergies and medications may be missed.
5. Emergency contacts may be difficult to reach.
6. Different organizations may use separate systems.
7. Manual processes increase the possibility of human error.
8. There may be delays in hospital preparation.

3.4 Major Gaps

The main gap is the **lack of a unified and secure platform for real-time emergency information sharing and coordination.**

The proposed system addresses this gap by connecting patients, first responders, ambulance teams, hospitals, and emergency contacts through a controlled software platform.

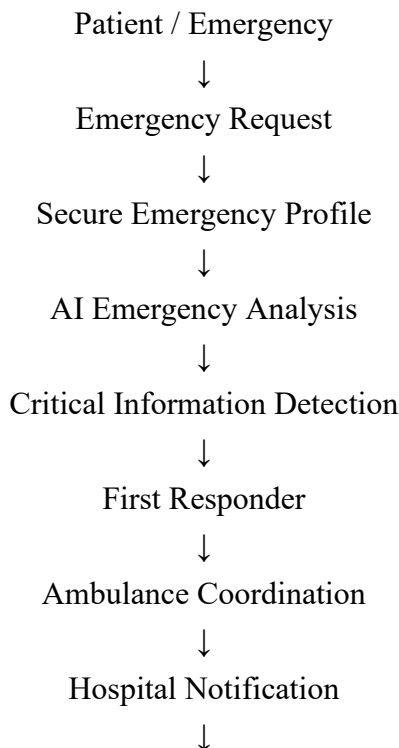
4. Proposed Software Solution and Software Engineering Principles

4.1 Proposed Solution

The proposed system is a **secure AI-based emergency coordination platform.**

When an emergency occurs, the system identifies the emergency, retrieves authorized patient information, generates a concise AI emergency summary, highlights critical medical information, shares the location with authorized responders, and sends appropriate notifications.

Basic System Flow



Emergency Treatment

4.2 Major Functional Modules

1. Emergency Profile Management

Stores:

- Patient details
- Blood group
- Allergies
- Medications
- Medical history
- Emergency contacts

2. AI Emergency Summary

AI analyzes available information and creates a short emergency summary that can be quickly understood by responders.

3. Critical Alert Module

Highlights important information such as:

ALLERGY → Penicillin

MEDICATION → Insulin

CRITICAL CONDITION → Asthma

4. First-Responder Module

Allows authorized responders to access relevant emergency information.

5. Location Sharing Module

Shares the emergency location with authorized responders and ambulance teams.

6. Hospital Notification Module

Sends emergency information to the selected hospital.

7. Emergency Contact Module

Notifies registered emergency contacts.

8. Authentication and Authorization Module

Ensures only authorized users can access protected information.

9. Emergency History Module

Maintains a record of emergency events and access activities.

10. Admin Module

Allows administrators to manage users, permissions, and system operations.

4.3 Functional Requirements

The system should:

1. Allow users to create and update emergency profiles.
2. Store medical and emergency information.
3. Authenticate users.
4. Generate AI-based emergency summaries.
5. Highlight critical medical information.
6. Share emergency location.
7. Notify first responders.
8. Notify hospitals.
9. Notify emergency contacts.
10. Maintain emergency and access history.
11. Provide role-based access.
12. Allow administrators to manage the system.

4.4 Non-Functional Requirements

Security

Patient information must be protected using authentication, authorization, encryption, and secure data handling.

Reliability

The system should continue functioning correctly during emergency situations.

Availability

The system should be available whenever an emergency occurs.

Usability

The interface should be simple because first responders have limited time.

Performance

Emergency information and alerts should be delivered quickly.

Scalability

The system should support increasing numbers of users, hospitals, and emergency events.

Maintainability

The software should be designed so that individual modules can be updated without affecting the complete system.

4.5 Software Engineering Principles

Principle	Application
Modularity	Separate modules are created for AI, users, alerts, location, hospitals, and security.
Scalability	Cloud-based architecture can support increasing users and emergency requests.

Principle	Application
Maintainability	Modular code makes testing, debugging, and updating easier.
Reliability	Backup, error handling, logging, and reliable notification mechanisms improve system reliability.
Usability	Simple dashboards and clear emergency alerts help responders work quickly.
Security	Authentication, authorization, encryption, and role-based access protect sensitive information.

5. Methodology / Framework Justification

Selected Methodology: Agile Scrum

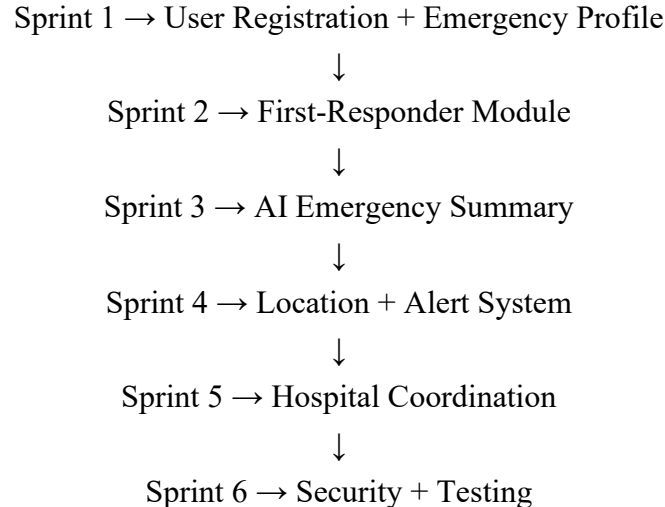
Agile Scrum is selected for developing the proposed system.

Why Agile Scrum?

The emergency response system contains multiple modules such as AI analysis, emergency profiles, location sharing, notifications, hospital coordination, and security. Developing all these components at once can increase project risk.

Agile allows the system to be developed **incrementally in small iterations called sprints**.

Example Sprint Plan



Advantages of Agile Scrum

1. Requirements can be changed based on stakeholder feedback.
2. Working software is delivered incrementally.
3. Critical modules can be developed first.
4. Testing is performed continuously.
5. Problems can be identified early.
6. First responders and hospital staff can provide feedback.

7. Security requirements can be reviewed throughout development.
8. The system can be improved after every sprint.

Scrum Roles

Product Owner: Represents emergency-service and user requirements.

Scrum Master: Ensures that the Scrum process is followed and removes development obstacles.

Development Team: Designs, develops, tests, and maintains the software.

Maintenance

After deployment, Agile supports continuous maintenance through:

User Feedback → New Requirement → Sprint Planning → Development → Testing → Release

Therefore, Agile Scrum is suitable because the system is **complex, user-centered, safety-critical, and likely to evolve based on real-world feedback.**

6. Expected Outcomes and Limitations

6.1 Expected Benefits

The proposed system is expected to:

1. Reduce emergency response delays.
2. Provide faster access to critical patient information.
3. Improve communication between responders and hospitals.
4. Reduce errors caused by manual information collection.
5. Highlight allergies and medications quickly.
6. Improve emergency location tracking.
7. Help hospitals prepare before the patient arrives.
8. Improve coordination among emergency stakeholders.
9. Protect sensitive patient information.
10. Improve overall emergency decision-making.

6.2 How the Solution Addresses Challenges

Existing Challenge	Proposed Solution
Missing medical information	Secure emergency profile
Slow information sharing	Real-time digital communication
Information overload	AI-generated emergency summary
Unknown emergency location	Location-sharing module
Allergy/medication missed	Critical alert module

Existing Challenge	Proposed Solution
Hospital not prepared	Advance hospital notification
Unauthorized access	Authentication and role-based authorization
Manual errors	Automated information processing

6.3 Risks and Implementation Challenges

1. Internet or network failure during emergencies.
2. Incorrect or outdated patient information.
3. AI may generate an incorrect or incomplete summary.
4. Unauthorized access or cyberattacks.
5. Data privacy concerns.
6. Integration difficulties with existing hospital systems.
7. Users may not update their emergency profiles regularly.
8. High availability and reliability are difficult to guarantee.
9. Training may be required for first responders and hospital staff.

6.4 Limitations

- The system depends on the availability of communication networks.
- AI recommendations should not replace professional medical judgment.
- The quality of results depends on the accuracy of stored patient information.
- Integration with different hospital systems may be difficult.
- Maintaining strong security and privacy requires continuous monitoring.

6.5 Future Enhancements

1. Integration with **smartwatches and wearable health devices**.
2. IoT-based automatic emergency detection.
3. Multilingual emergency voice assistant.
4. Integration with hospital electronic health records.
5. AI-based emergency risk prediction.
6. Automatic ambulance route optimization.
7. Offline emergency information access.
8. Advanced fraud and unauthorized-access detection.
9. Integration with smart city emergency services.
10. Predictive analytics for emergency response planning.

Conclusion

The **AI-Based Emergency Response and First-Responder Coordination System** provides a centralized, secure, and intelligent platform for improving emergency communication and coordination. It addresses major limitations of manual emergency processes by providing quick access to critical information, AI-based summarization, location sharing, alerts, and hospital coordination.

The use of **Agile Scrum** allows the system to be developed incrementally, tested continuously, and improved using feedback from patients, first responders, ambulance teams, and hospitals. By applying software engineering principles such as **modularity, scalability, maintainability, reliability, usability, and security**, the proposed system can provide a practical and extensible solution for modern emergency response.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability,	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

	maintainability, security, etc.).			
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25